

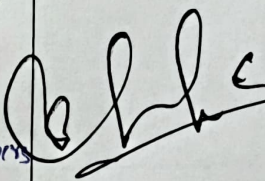
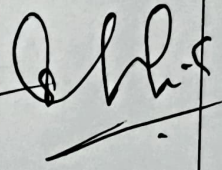
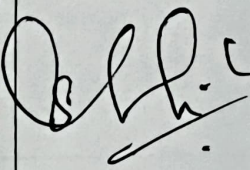
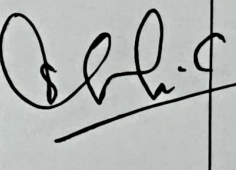
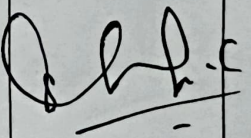
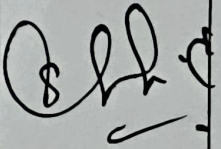
**Department of Computer Science & Engineering**  
 Global Campus, Jakkasandra Post, Kanakapura Taluk, Ramanagara District, Pin Code: 562 112

**FACULTY OF ENGINEERING AND TECHNOLOGY**  
**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**  
**ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING**

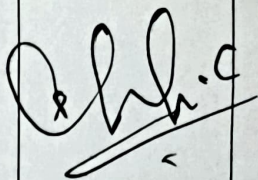
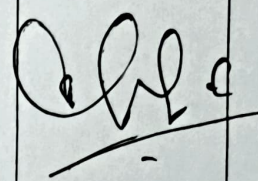
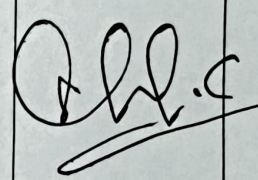
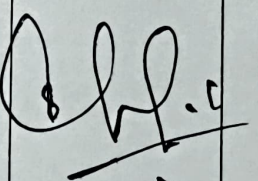
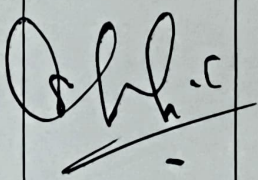
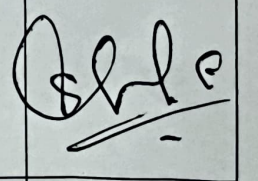
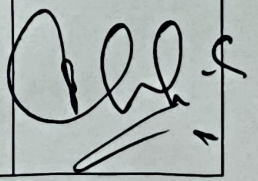
**PCL Attendance**

**Team No. : CoEPCL- 0216**

**Project Title : SMART WASTE MANAGEMENT ( ECOSTREAM-AI )**

| Sl. No. | Date of the Meeting            | Points Discussed                                                                                                                                 | Members Absent (Roll No)                                                             | Signature of the Mentor                                                               |
|---------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1.      | 9 <sup>th</sup> /Jan.<br>2026. | Discussed project idea with guide and finalized the problem statement. Roles and responsibilities were assigned among team members.              |   |    |
| 2.      | 13/01/26.                      | Updated guide on development environment setup and Github repository Creation. discussed project architecture and required technologies.         |  |  |
| 3.      | 22/01/26.                      | Presented Collected datasets and municipal PDF Sources to the guide. Received Suggestions regarding dataset pre processing and train-validation. |                                                                                      |  |
| 4.      | 27/01/26.                      | Demonstrated literature review progress and GAN-based augmentation approach. Discussed implementation strategy for mixed waste image generation. |                                                                                      |  |
| 5.      | 9/02/26.                       | Showed docker based multi service setup integrating FastAPI, Yolo, PostgreSQL, and Chroma DB. Guide reviewed backend structure.                  |                                                                                      |  |



|     |           |                                                                                                                                                    |                                                                                       |
|-----|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 6.  | 19/02/26. | Presented YOLOv11 Segmentation model training progress and baseline comparison results. discussed detection accuracy and API response structure    |    |
| 7.  | 25/02/26. | Demonstrated RAG Pipeline implementation using chroma DB and Langchain. Guide reviewed disposal advice generation workflow and document ingestion. |    |
| 8.  | 6/03/26.  | Presented backend orchestration workflow integrating Yolo v11 and postgres SQL logging. Discussed API endpoints and end to end testing results.    |    |
| 9.  | 12/03/26. | demonstrated frontend Citizen portal with Camera Capture GPS integration Chatbot interface Guide reviewed user interface design.                   |    |
| 10. | 21/03/26  | Presented Government dashboard with heatmap visualization and collection management features.                                                      |  |
| 11. | 3/04/26.  | Updated guide on predictive analytics module and Random forest hotspot prediction. Discussed IEEE paper drafting KPI Evaluation & documentation.   |  |
| 12. | 11/04/26. | demonstrated complete deployed system with frontend backend Yolo detection RAG. advice heatmap and prediction module.                              |  |